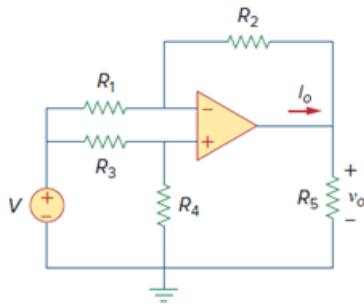


Problem

Using Fig. 5.50, design a problem to help other students better understand how ideal op amps work.

Figure. 5.50:



Step-by-step solution

Step 1 of 7

Refer to Figure 5.50 in the textbook.

Draw the modified circuit diagram as shown in Figure 1.

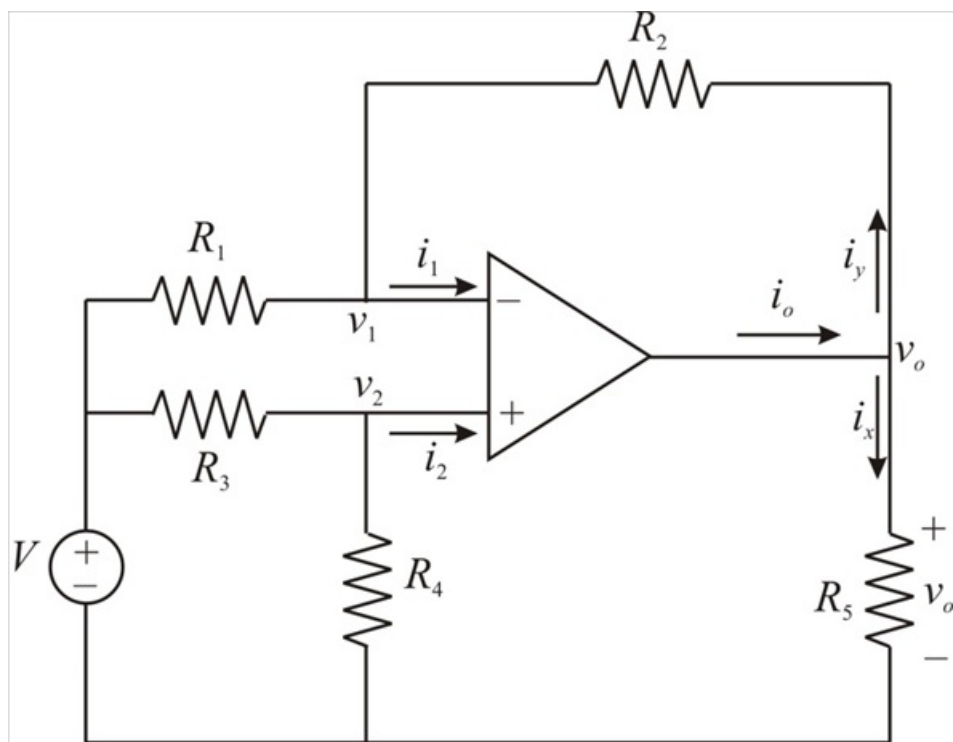


Figure 1

Step 2 of 7

Determine the voltage across v_2 .

Apply Kirchhoff's current law at node v_2 .

$$\frac{v_2 - V}{R_3} + \frac{v_2}{R_4} = 0$$

$$\frac{R_4(v_2 - V) + R_3v_2}{R_3R_4} = 0$$

$$R_4(v_2 - V) + R_3v_2 = 0$$

$$R_4v_2 - R_4V + R_3v_2 = 0$$

$$(R_3 + R_4)v_2 - R_4V = 0$$

$$(R_3 + R_4)v_2 = R_4V$$

$$v_2 = \left(\frac{R_4V}{R_3 + R_4} \right)$$

Step 3 of 7

Consider in op-amp the potential difference between non-inverting terminal and inverting terminal is zero.

$$v_2 - v_1 = 0$$

$$v_2 = v_1$$

Consider voltage at v_2 is,

$$v_2 = \left(\frac{R_4V}{R_3 + R_4} \right)$$

$$v_1 = \left(\frac{R_4V}{R_3 + R_4} \right)$$

Therefore, the voltage at inverting terminal v_1 is $\left(\frac{R_4}{R_3 + R_4} \right)$.

Step 4 of 7

Apply Kirchhoff's current law at node v_1 .

$$\frac{v_1 - V}{R_1} + \frac{v_1 - v_o}{R_2} = 0$$

$$\frac{R_2(v_1 - V) + R_1(v_1 - v_o)}{R_1R_2} = 0$$

$$R_2(v_1 - V) + R_1(v_1 - v_o) = 0$$

$$R_2v_1 - R_2V + R_1v_1 - R_1v_o = 0$$

$$(R_1 + R_2)v_1 - R_2V - R_1v_o = 0$$

$$R_1v_o = (R_1 + R_2)v_1 - R_2V$$

$$v_o = \frac{(R_1 + R_2)v_1}{R_1} - \frac{R_2}{R_1}V \dots\dots (1)$$

Step 5 of 7

Substitute $\left(\frac{R_4}{R_3 + R_4}\right)V$ for v_1 in equation 1.

$$\begin{aligned} v_o &= \frac{(R_1 + R_2) \left[\left(\frac{R_4}{R_3 + R_4} \right) V \right]}{R_1} - \frac{R_2}{R_1} V \\ &= \frac{(R_1 + R_2) R_4 V}{(R_3 + R_4) R_1} - \frac{R_2}{R_1} V \\ &= V \left[\frac{R_1 R_4 + R_2 R_4 - R_2 R_3 - R_2 R_4}{(R_3 + R_4) R_1} \right] \\ v_o &= V \left[\frac{R_1 R_4 - R_2 R_3}{(R_3 + R_4) R_1} \right] \end{aligned}$$

Therefore, the output voltage v_o is $V \left(\frac{R_1 R_4 - R_2 R_3}{R_1 (R_3 + R_4)} \right) V$.

Step 6 of 7

Determine the output current.

Apply Kirchhoff's current law at the output node.

$$\begin{aligned} i_o &= i_x + i_y \\ &= \left(\frac{v_o}{R_5} \right) + \left(\frac{v_o - v_1}{R_2} \right) \\ &= v_o \left(\frac{1}{R_5} + \frac{1}{R_2} \right) - \left(\frac{v_1}{R_2} \right) \\ i_o &= v_o \left(\frac{R_2 + R_5}{R_2 R_5} \right) - v_1 \left(\frac{1}{R_2} \right) \dots\dots (2) \end{aligned}$$

Step 7 of 7

Substitute $V \left[\frac{R_1 R_4 - R_2 R_3}{(R_3 + R_4) R_1} \right]$ for v_o , and $\left(\frac{R_4 V}{R_3 + R_4} \right)$ for v_1 in equation 2.

$$\begin{aligned}
 i_o &= \left[\left(\frac{R_1 R_4 - R_2 R_3}{R_1 (R_3 + R_4)} \right) V \right] \left(\frac{R_2 + R_5}{R_2 R_5} \right) - \left[\left(\frac{R_4 V}{R_3 + R_4} \right) V \right] \left(\frac{1}{R_2} \right) \\
 &= V \left[\left(\frac{R_1 R_4 - R_2 R_3}{R_1 (R_3 + R_4)} \right) \left(\frac{R_2 + R_5}{R_2 R_5} \right) - \left(\frac{R_4}{R_2 (R_3 + R_4)} \right) \right] \\
 &= \left(\frac{V}{R_2 (R_3 + R_4)} \right) \left[\left(\frac{(R_1 R_4 - R_2 R_3)(R_2 + R_5)}{R_1 R_5} \right) - R_4 \right] \\
 &= \left(\frac{V}{R_2 (R_3 + R_4)} \right) \left[\frac{R_1 R_2 R_4 - R_2^2 R_3 + R_1 R_4 R_5 - R_2 R_3 R_5 - R_1 R_4 R_5}{R_1 R_5} \right] \\
 &= \left(\frac{V}{R_2 (R_3 + R_4)} \right) \left[\frac{R_1 R_2 R_4 - R_2^2 R_3 - R_2 R_3 R_5}{R_1 R_5} \right] \\
 &= \left(\frac{V}{R_3 + R_4} \right) \left[\frac{R_1 R_4 - R_2 R_3 - R_3 R_5}{R_1 R_5} \right] \\
 i_o &= \left(\frac{R_1 R_4 - R_2 R_3 - R_3 R_5}{R_1 R_5 (R_3 + R_4)} \right) V
 \end{aligned}$$

Therefore, the output current i_o is $\boxed{\left(\frac{R_1 R_4 - R_2 R_3 - R_3 R_5}{R_1 R_5 (R_3 + R_4)} \right) V \text{ A}}.$